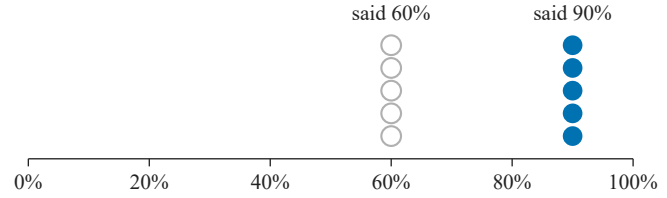
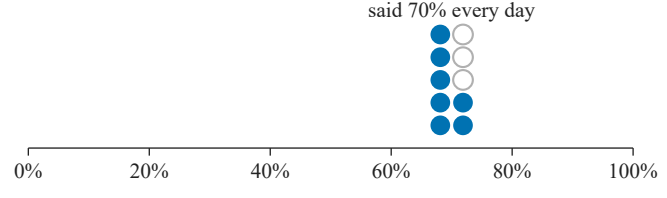
Forecaster A: perfect ranking, wrong numbers



Every rainy day got a higher chance than every dry day: **ranking is perfect** (AUC = 1).

But on the days it said 60%, it never rained; on the days it said 90%, it always did: **the numbers cannot be taken at face value.**

Forecaster B: right numbers, no ranking



On the days it said 70%, it rained on 7 of 10: **the numbers are right** (calibrated).

But it gave every day the same chance, so it **cannot tell the days apart** (AUC = 0.5).

a day it rained a dry day horizontal position = the forecast chance of rain